



JERUSALEM COLLEGE OF ENGINEERING, CHENNAI – 600 100
(An Autonomous Institution Affiliated to Anna University, Chennai)

Reg. No. :

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Question Paper Code : 215024

B.E./B.Tech. DEGREE END SEMESTER EXAMINATIONS, APRIL/MAY 2025

Fourth Semester

Computer Science and Engineering

JCS2401 – DATABASE MANAGEMENT SYSTEMS

(Common to Computer Science and Engineering (Artificial Intelligence and Machine Learning), Computer Science and Engineering (Cyber Security), Artificial Intelligence and Data Science, Computer Science and Business Systems, and Information Technology)

(Regulation 2023)

Time : Three hours

Maximum : 100 marks

Answer ALL questions

PART A – (10 × 2 = 20 marks)

1. Define database management system?
2. What is embedded SQL?
3. List the various types of attributes in ER model.
4. Define Primary Key.
5. What is serializability? Give its various forms.
6. Mention the four conditions for deadlock.
7. Define rotational latency time.
8. Define Query optimization
9. Define Object-Oriented Database Management System.
10. List the benefits of XML Schema over DTD.

PART B – (5 × 16 = 80 marks)

11. (a) List out the operations of the relational algebra and explain with suitable examples.

Or

(b) Discuss in detail about Embedded SQL and Dynamic SQL with necessary examples.

12. (a) Explain in detail the various types of Functional dependency.

Or

(b) Discuss in detail about the following.

(i) Fourth Normal Form (4)

(ii) Multivalued Dependency (4)

(iii) Fifth Normal Form (4)

(iv) Join dependency and lossless decomposition (4)

13. (a) What is transaction? Explain in detail about the various states of a transaction and ACID property.

Or

(b) Consider the following schedules. The actions are listed in the order they are scheduled, and pre-fixed with the transaction name.

S1: T1: R(X), T2: R(X), T1: W(Y), T2: W(Y), T1: R(X), T2: R(Y)

S2: T3: W(X), T1: R(X), T1: W(Y), T2: R(Z), T2: W(Z), T3: R(Z)

Draw the precedence graph and check whether the schedule is Conflict or View serializable and also define all relevant equivalent serial schedule.

14. (a) Illustrate the various types of Hashing, the techniques and issues for each with appropriate examples.

Or

(b) List and explain with example the General transformation rules /Equivalence rules for Relational algebra expressions.

15. (a) Illustrate the Object Database Concepts, Object Relational Features.

Or

(b) Describe the implementation and usage of Spatial Database.